

Novelty Summary

The proposed **Telemetry-Aware Multi-Agent Infrastructure-as-Code Assurance System for Smart Manufacturing Cloud Environments** introduces an intelligent framework that integrates Infrastructure-as-Code validation, security assessment, deployment verification, configuration drift detection, and runtime telemetry analysis within a single coordinated workflow. Unlike existing studies that focus on isolated tasks such as Terraform generation, Kubernetes validation, PLC verification, or Industrial IoT security, the proposed framework provides a comprehensive, cloud-native solution for validating Infrastructure-as-Code across smart-manufacturing environments.

Novel Features

1. Unified Multi-Agent Validation Framework

- Integrates specialized LLM agents for IaC parsing, syntax validation, security assessment, deployment verification, configuration drift detection, cost analysis, and recommendation generation within one intelligent workflow.

2. Support for Multiple Infrastructure-as-Code Technologies

- Validates Terraform, AWS CloudFormation, AWS CDK, Kubernetes manifests, and AWS Serverless Application Model (SAM), overcoming the single-platform limitations of existing research.

3. Telemetry-Aware Infrastructure Validation

- Combines static Infrastructure-as-Code analysis with runtime telemetry collected from AWS CloudWatch, deployment logs, and cloud resource monitoring to improve validation accuracy and detect configuration drift.

4. Cloud-Native AWS Architecture

- Leverages AWS services including Amazon Bedrock, AWS Lambda, Amazon CloudWatch, Amazon S3, Amazon API Gateway, Amazon Cognito, Amazon DynamoDB, Amazon RDS, Amazon SNS, and AWS IAM to provide a scalable, secure, and serverless validation platform.

5. Enhanced Security and Automation

- Performs automated detection of security misconfigurations, validates deployment readiness, and generates explainable remediation recommendations, significantly reducing manual review effort.

6. Improved Validation Accuracy and Reliability

- Integrates intelligent reasoning with deployment verification and runtime monitoring to provide more reliable Infrastructure-as-Code validation than approaches based solely on static analysis or benchmark evaluation.

7. Scalable Smart-Manufacturing Deployment

- The modular architecture enables seamless integration with cloud-native manufacturing systems and can be extended to additional Infrastructure-as-Code technologies, industrial applications, and enterprise cloud environments with minimal modification.